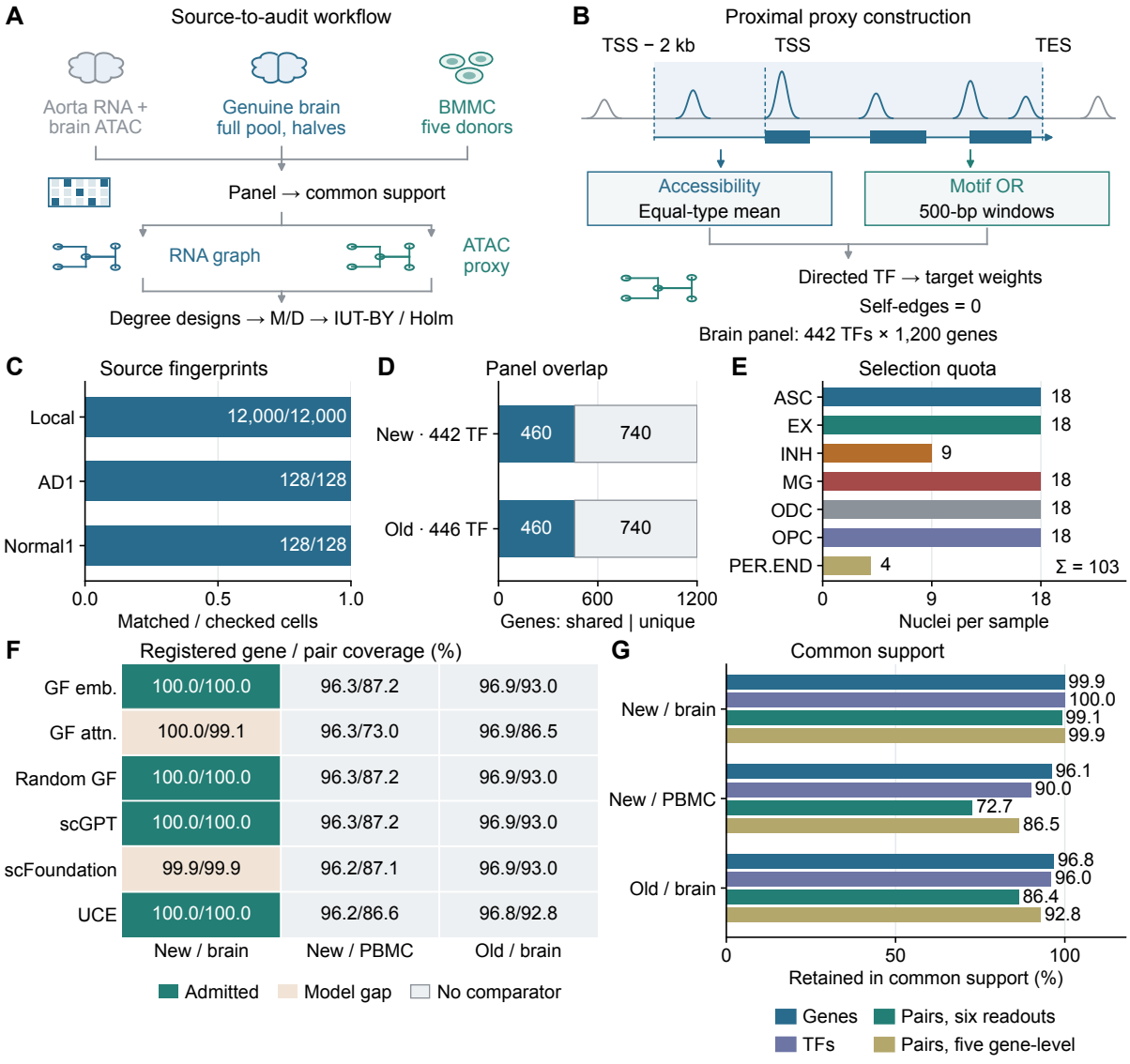
A
Source-to-audit workflow
Aorta RNA + brain ATAC
Genuine brain full pool, halves
BMMC five donors
Panel → common support
RNA graph
ATAC proxy
Degree designs → M/D → IUT-BY / Holm
B
Proximal proxy construction
TSS − 2 kb
TSS
TES
Accessibility
Equal-type mean
Motif OR
500-bp windows
Directed TF → target weights
Self-edges = 0
Brain panel: 442 TFs × 1,200 genes
C
Source fingerprints
Local
AD1
Normal1
12,000/12,000
128/128
128/128
0.0
0.5
1.0
Matched / checked cells
D
Panel overlap
New · 442 TF
Old · 446 TF
460
740
460
740
0
600
1200
Genes: shared | unique
E
Selection quota
ASC
EX
INH
MG
ODC
OPC
PER.END
18
18
9
18
18
18
4
Σ = 103
0
9
18
Nuclei per sample
F
Registered gene / pair coverage (%)
GF emb.
GF attn.
Random GF
scGPT
scFoundation
UCE
100.0/100.0
100.0/99.1
100.0/100.0
100.0/100.0
99.9/99.9
100.0/100.0
96.3/87.2
96.3/73.0
96.3/87.2
96.3/87.2
96.2/87.1
96.2/86.6
96.9/93.0
96.9/86.5
96.9/93.0
96.9/93.0
96.9/93.0
96.8/92.8
New / brain
New / PBMC
Old / brain
Admitted
Model gap
No comparator
G
Common support
New / brain
New / PBMC
Old / brain
99.9
100.0
99.1
99.9
96.1
90.0
72.7
86.5
96.8
96.0
86.4
92.8
0
50
100
Retained in common support (%)
Genes
Pairs, six readouts
TFs
Pairs, five gene-level